Data-driven Wildfire Frequency Prediction for Real-Estate Applications

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Abstract

Traditional wildfire burn probability assessment relies heavily on expensive physical fire spread simulations, static fuel cover, and outdated historical ignition data. In partnership with Cotality, this capstone project develops a scalable statistical framework for estimating wildfire burn probability across California using publicly available dynamic geospatial, vegetation, topographic, and weather data. Using historical wildfire perimeter data from the Monitoring Trends in Burn Severity (MTBS) program for California, we replicated and adapted the statistical methods presented in Anderegg et al. (2022). Building on last quarter's baseline classification models, we expanded the spatial domain to California and implemented gradient-boosted decision trees (XGBoost) with hyperparameter optimization using Optuna. Unlike previous models that struggled under severe class imbalance, our current framework emphasizes precision–recall AUC (PR-AUC), supports threshold tuning, and leverages GPU acceleration for scalability. Results on random test splits show improved minority-class detection and stronger capture of nonlinear relationships, while temporal holdout results highlight remaining challenges in generalization. Overall, this framework provides a cost-efficient, reproducible approach for spatial burn-probability modeling and offers a foundation for future wildfire risk assessment and resilience planning.

Code: https://github.com/lachuck331/CotalityQ1Capstone

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1 Introduction

Wildfires pose a growing threat across California, where rising temperatures, prolonged drought, shifting fuel conditions, and diverse fire-prone landscapes create increasingly favorable conditions for ignition and fire spread. These risks extend beyond any single county, affecting communities across a wide range of climatic and ecological zones and creating growing demand for accurate, scalable, and timely burn-probability assessment. However, many operational wildfire models still rely on computationally intensive physical fire spread simulations that require detailed fuel maps, calibrated ignition inputs, and assumptions that may be difficult to maintain under changing vegetation and climate conditions. In practice, these approaches can be costly, inflexible, and difficult to scale for broad resilience planning. As a result, there is growing interest in alternative risk-modeling frameworks that use publicly available environmental data to estimate wildfire likelihood in a reproducible and cost-efficient way. This capstone project, conducted in collaboration with Cotality, a risk analytics organization focused on climate and catastrophe modeling, develops a data-driven wildfire burn-probability pipeline using publicly available environmental datasets rather than simulation-based fire spread models. Building on last quarter's county-level analysis in San Diego, this quarter expands the spatial domain to the full state of California and advances the modeling framework from baseline statistical classifiers to gradient-boosted methods designed for rare-event prediction.

1.1 Literature Review and Prior Work

Recent research has demonstrated that statistical models using climate and vegetation variables can successfully predict large-scale fire occurrence without the need for physical fire behavior simulations. A key example is the work of Anderegg et al. (2022), who developed a nationwide statistical fire model using gridded climate variables, vegetation attributes, and historical MTBS wildfire perimeters. Their model employed a hurdle regression framework, first classifying whether a fire occurs, then predicting burn area given a fire, and achieved high predictive power across U.S. forested landscapes. Importantly, their results showed that climate-driven variables such as temperature, precipitation, and climatic water deficit were strong, consistent predictors of fire activity. This demonstrated that statistical methods grounded in widely available datasets can replicate major wildfire patterns across large, diverse ecosystems.

Building on this foundation, Di Giuseppe et al. (2025) have proposed a global data-driven prediction system that shifts the focus from traditional fire danger ratings to predicting actual fire activity. This methodology represents a strategic departure from classical simulation-based models, such as FARSITE or FlamMap, which rely on computationally intensive physics engines and often result in false alarms in fuel-limited biomes by over-predicting danger in barren areas (Di Giuseppe et al. 2025, pp. 14, 183). To address the challenges of modeling catastrophic, "rare-event" occurrences within extremely imbalanced datasets, their framework utilizes **gradient-boosted decision trees** (implemented via the **XGBoost** library). Their results indicate that an intermediate-complexity machine learning architecture can

achieve superior performance by effectively learning the non-linear conditions where fire is inhibited or fueled through integrated weather, fuel, and ignition data (Di Giuseppe et al. 2025, pp. 15, 161, 522).

The efficacy of such data-driven modeling was notably demonstrated during the 2025 Los Angeles wildfires, where the model successfully localized ignitions by accounting for anomalous fuel accumulation ("hydroclimate whiplash") that traditional indices failed to detect (Di Giuseppe et al. 2025, pp. 350, 353, 358). This project extends these recent advancements to California by examining whether a similar climate and vegetation-driven statistical model can capture historical burn probability patterns at a high **800-meter resolution**. Developed in partnership with Cotality, this work aims to translate these probabilities into actionable intelligence for **insurance and mortgage pricing**, while simultaneously providing high-fidelity data for **public safety** and **urban planning**. By focusing specifically on burn probability, this work builds a bridge between global modeling frameworks and actionable community-level tools.

1.2 Relevant Data

This project integrates several categories of publicly available data that together characterize the climatic, ecological, and physical landscape of California. Monthly PRISM climate data provide precipitation, maximum temperature, vapor pressure deficit, and dewpoint temperature at an 800-meter resolution, offering detailed insight into the atmospheric conditions that precede wildfire activity. Vegetation dynamics are captured through MODIS NDVI, which serves as a proxy for fuel moisture and greenness, and annual NLCD land cover classifications, which identify dominant fuel types (e.g., shrubland, grassland, developed areas) and are upscaled to match the climate data grid. Topographic variables such as elevation, slope, and aspect are derived from USGS 1-arc-second digital elevation models and repeated across monthly time steps to align with the climate and vegetation datasets. Historical wildfire perimeters from the MTBS dataset form the target variable, enabling the construction of a monthly binary classification task in which each grid cell is labeled as burned or unburned. All datasets are carefully regridded and aligned to a shared spatial resolution and domain extent, following the data preparation instructions outlined in the project guidelines.

2 Methods

2.1 Data Processing

The data processing pipeline integrates multiple environmental datasets to create a unified spatiotemporal dataset for wildfire prediction. In this iteration, the spatial domain has been expanded from San Diego County to the entire state of California, and the temporal range covers January 2000 to December 2024.

2.1.1 Input Data Layers

- **Climate:** Monthly PRISM climate variables at 800m spatial resolution (January 2000 to December 2024):
  - Precipitation (`ppt`)
  - Maximum Temperature (`tmax`)
  - Maximum Vapor Pressure Deficit (`vpdmax`)
  - Mean Dewpoint Temperature (`tdmean`)
- **Topography:** USGS 1 Arc Second Digital Elevation Models (DEM).
- **Vegetation:** MODIS/Terra Vegetation Indices Monthly L3 Global 1km SIN Grid V061 (February 2000 to December 2024).
- **Land Cover:** Annual NLCD (National Land Cover Database) land cover data (2000 to 2024).
- **Target:** MTBS (Monitoring Trends in Burn Severity) Fire Perimeter dataset.
- **Spatial Domain:** Polygon shapefile of the state of California.

2.1.2 Pre-processing Steps

All input data layers were pre-processed to a consistent resolution (800m PRISM grid) and domain extent (California).

Climate (PRISM).

- Subset monthly PRISM climate variables to the domain fully encompassing California.

Topography (DEM).

- Slope and aspect were calculated from the high-resolution 1 arc second DEM.
- Elevation was aggregated to the PRISM reference grid, while slope and aspect were derived from DEM gradients and then aggregated to the same grid.
- These static layers were replicated to repeat monthly, matching the time steps of the climate data.

Land Cover (NLCD).

- For each year, NLCD classes were **upscaled** to the 800m PRISM grid by calculating the **dominant class** (mode) for each 800m grid cell.
- Annual files were replicated to repeat monthly, aligning with the climate data time steps.

Vegetation (NDVI).

- Monthly MODIS/Terra NDVI data were regridded to the PRISM 800m grid using a **nearest neighbor** algorithm.

2.1.3 Target Variable Generation (MTBS)

The target variable, `burned_area`, was generated by rasterizing wildfire perimeters.

Filtering. The MTBS dataset was subset to events that:

- Occurred in California.
- Occurred between 2000 and 2024.
- Were labeled specifically as "Wildfire" (excluding prescribed burns, etc.).

Rasterization. For each month in the time series:

- Events within the spatial domain were identified.
- Perimeters were rasterized to the 800m grid.
- Monthly binary wildfire labels were created by rasterizing MTBS wildfire perimeters for fires whose ignition date fell in each month.

2.1.4 Final Integration and Cleaning

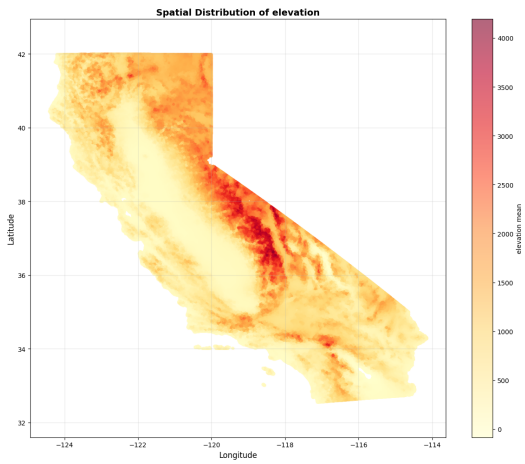
- All processed layers were merged into a single dataset.
- **Missing Data Handling:**
  1. Rows with missing values in climate variables (`ppt`, `tmax`, etc.) or NDVI columns were dropped to ensure data quality.
  2. The first month of the time series (January 2000) was removed for all locations because the first month of NDVI coverage was in February 2000.
  3. Data were subset to only include coordinates that are present in all input data layers.
  4. Rows with missing NLCD data (`land_cover`) after February 2000 were also excluded during the encoding process.
- **Splitting:**
  - **Validation Set:** Data from years 2005–2009 were set aside for final validation.
  - **Train/Test Split:** The remaining data (years < 2005 and > 2009) were split into training (70%) and testing (30%) sets.
- **Scaling:** Features were standardized using `StandardScaler` to ensure zero mean and unit variance, facilitating convergence for baseline linear models.

2.2 Exploratory Data Analysis (EDA)

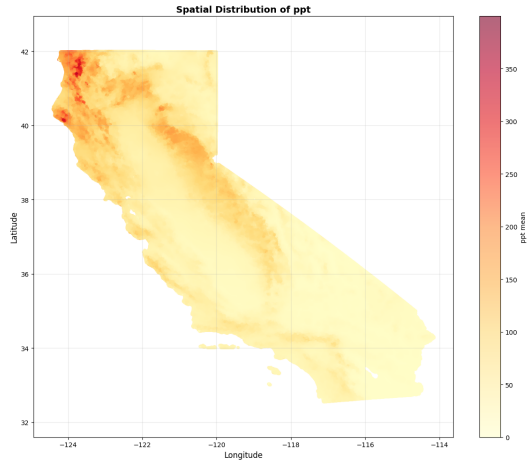
An extensive EDA was conducted to understand data quality, distributions, and relationships across the expanded California domain.

2.2.1 Dataset Overview

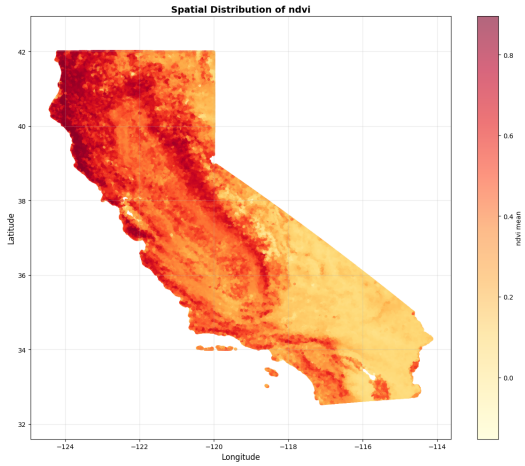
- **Initial Shape:** Approximately 424 million rows and 14 columns, reflecting the statewide coverage.
- **Temporal Coverage:** January 2000 to December 2024.
- **Memory Usage:** Approximately 40 GB, necessitating efficient storage formats like Parquet and streaming processing via Polars.



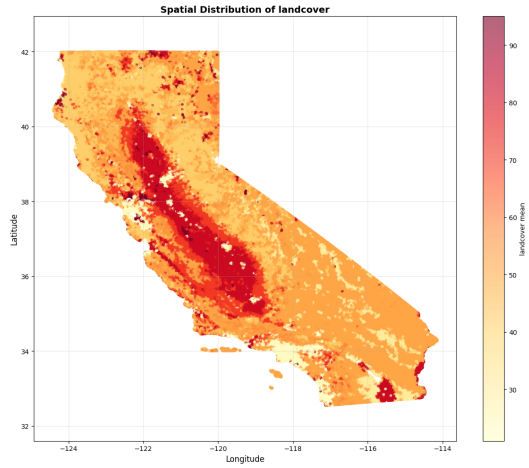
(a) Elevation



(b) Precipitation (`ppt`)



(c) NDVI



(d) Land cover

Figure 1: Representative spatial distributions of key input variables across the California study domain after alignment to the common 800m grid. These maps illustrate the substantial geographic heterogeneity in terrain, moisture, vegetation, and land cover that the model uses for wildfire prediction.

2.2.2 Missing Data Analysis

Extent. Significant missingness was observed in key predictors due to the expanded bounding box covering ocean and non-domain areas:

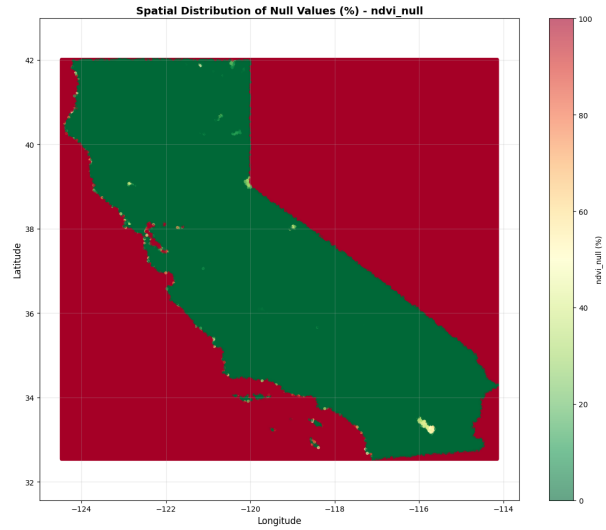
- **Climate** (`ppt`, `tmax`, `vpdmax`, `tdmean`): Approximately 238 million missing values (approx. 56% of the raw grid), primarily corresponding to ocean/coastline areas outside the valid PRISM domain.
- **NDVI & Land Cover:** Similar high magnitudes of missingness were observed where satellite coverage did not overlap with the grid.

Patterns.

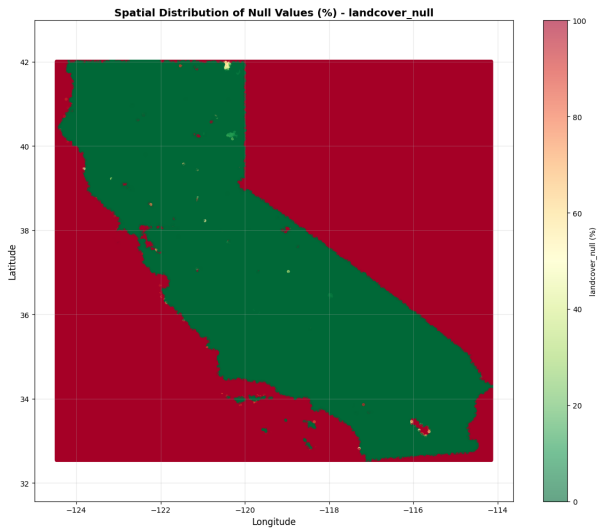
- **Temporal Anomaly:** January 2000 showed 100% missing NDVI data due to the MODIS satellite launch timeline. This month was dropped.
- **Spatial Missingness:** Climate and land-cover nulls were found to be spatially clustered, corresponding to ocean/coastline areas.

Implication. The non-random nature of missingness (representing non-land areas) justified the decision to drop incomplete rows to isolate the valid land-based spatial model.

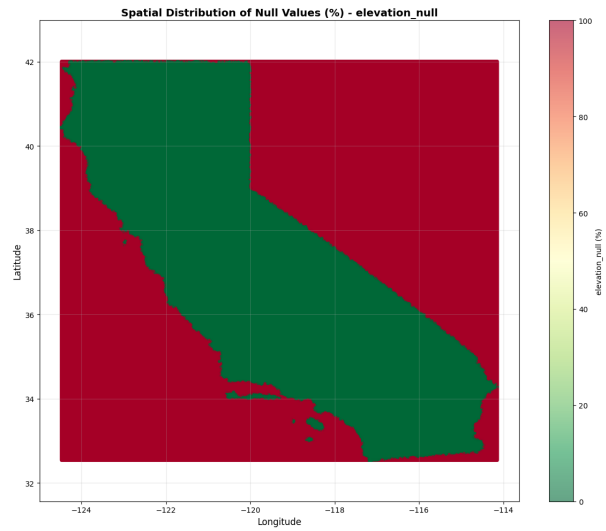
(a) PPT: nulls



(b) NDVI: nulls



(c) Landcover: nulls



(d) Elevation: nulls

Figure 2: Spatial distribution of missingness across major input layers. Missing values are concentrated primarily outside valid land-domain coverage, especially in ocean and boundary regions, rather than being randomly distributed across California.

2.2.3 Variable Distributions

- **Precipitation (`ppt`):** Highly right-skewed, reflecting the dry climate of California.
- **Dew Point Temperature (`tdmean`):** Approximately normally distributed.
- **Maximum Temperature (`tmax`):** Slightly right-skewed but largely normal.
- **Target (`burned_area`):** Extreme class imbalance persists in the statewide dataset, confirmed by distribution analysis.

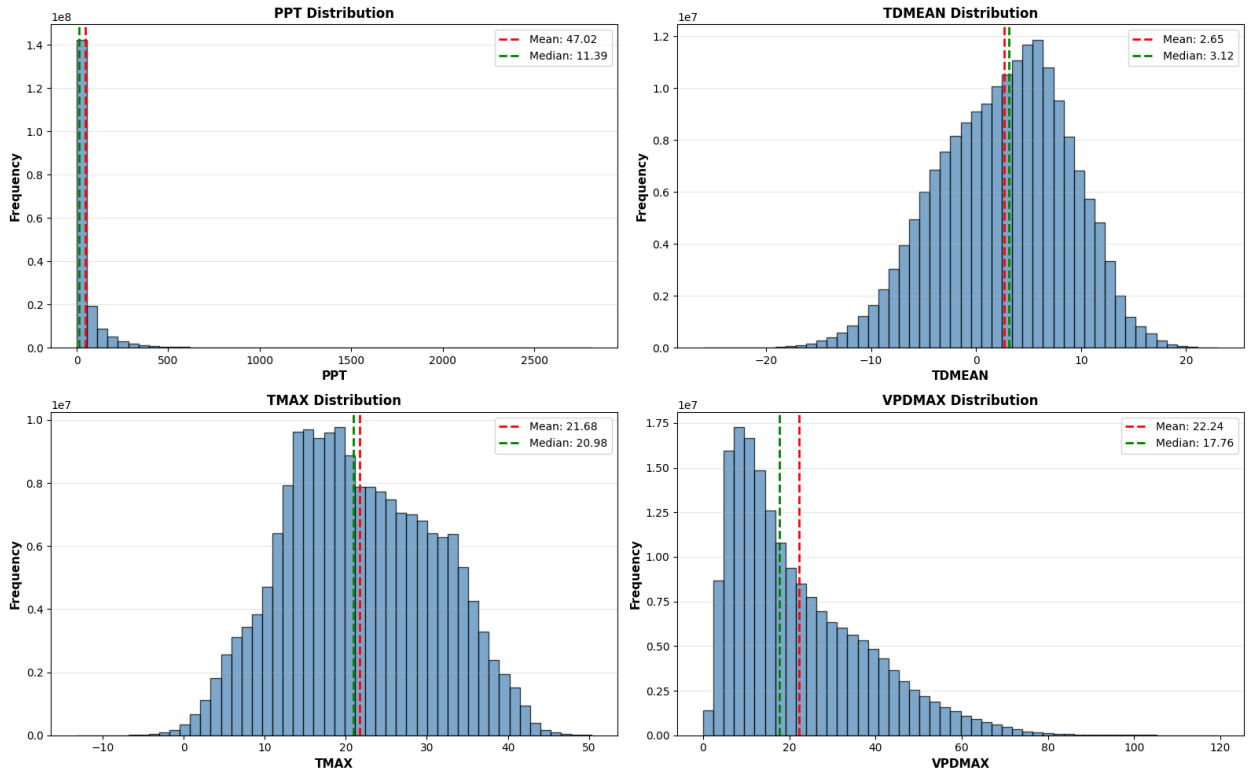


Figure 3: Distributions of the main climate variables used in modeling: precipitation (`ppt`), dew point temperature (`tdmean`), maximum temperature (`tmax`), and vapor pressure deficit (`vpdmax`). Precipitation and vapor pressure deficit are visibly right-skewed, supporting the use of transformed features during modeling.

2.2.4 EDA Insights

- **Land Cover:** Categorical variable requiring one-hot encoding.
- **Target:** Converted to `UInt8` type for memory-efficient classification.
- **Scaling:** Skewed distributions in `ppt` and `vpdmax` suggested the need for log-transformation or robust scaling.
- **Multicollinearity:** High correlation between `tmax` and `vpdmax` suggested the need for dimensionality reduction.

2.2.5 Feature Engineering

- **Lagging:** The target variable `burned_area` was "lagged" to predict future fires based on past conditions.
- **Log Transformation:** Log transformations were applied to the `ppt` and `vpdmax` variables.
- **Dimensionality Reduction:** PCA was applied to `tmax` and `vpdmax_log` to create a combined feature `tmax_vpdmax_log_pca`.

2.3 Model Selection

To address the increased data volume and severe class imbalance of the statewide dataset, we evolved our modeling strategy from linear baselines to gradient-boosted trees.

2.3.1 Baseline Models

Last quarter, we evaluated three classification models on the San Diego subdomain to establish performance benchmarks:

1. **Logistic Regression:** Employed for its interpretability and computational efficiency.
2. **Random Forest:** Tested for its ability to capture non-linear interactions between climate and vegetation.
3. **Linear SVM:** Evaluated for effectiveness in high-dimensional feature spaces.

While useful for initial insights, these models struggled to balance sensitivity and precision given the 0.1% positive class prevalence.

2.3.2 Primary Model: XGBoost

For the full California dataset, we implemented **XGBoost (Extreme Gradient Boosting)**, a scalable tree-boosting system. XGBoost was selected for several key advantages:

- **GPU Acceleration:** We utilized the `device="cuda"` parameter to accelerate training on the 400M+ row dataset, significantly reducing computation time compared to CPU-bound Scikit-Learn implementations.
- **Rare Event Handling:** Gradient boosting iteratively corrects errors from previous trees, allowing the model to focus on hard-to-classify minority samples (fire events) more effectively than bagging methods like Random Forest.
- **Performance Metrics:** Unlike the baseline accuracy metrics, the XGBoost implementation was optimized and evaluated using **Precision-Recall AUC (PR-AUC)** and ROC-AUC to explicitly account for the extreme class imbalance.

Hyperparameter optimization was conducted using Optuna to tune learning rate, tree depth, and subsampling ratios, further enhancing the model's ability to discriminate between burned and unburned grid cells.

Table 1: Baseline model performance on the test set for the San Diego Subdomain.

Model	Prec (Fire)	Prec (No Fire)	Rec (Fire)	Rec (No Fire)	F1 (Fire)	F1 (No Fire)	ROC-AUC	PR-AUC
Logistic Regression	0.01	1.00	0.87	0.84	0.01	0.91	0.9287	0.0154
Random Forest	0.00	1.00	0.00	1.00	0.00	1.00	0.5325	0.0061
Linear SVM	0.00	1.00	0.00	1.00	0.00	1.00	0.9114	0.0468

3 Initial Results

3.1 Baseline Model Performance Comparison

3.1.1 Logistic Regression

Logistic Regression performs the best among the three models on the rare-event task. Although precision remains low due to extreme imbalance, it achieves:

- **High recall (0.87)**, meaning it successfully identifies most months with fire.
- **Strong ROC-AUC (0.93)**, implying excellent separability between burned and non-burned cases.
- **Highest PR-AUC among the tested models**, though still low (0.015).

The model successfully leverages continuous climate and vegetation predictors and benefits from class-weighting, indicating that linear decision boundaries capture meaningful fire–environment relationships.

3.1.2 Random Forest

The Random Forest model appears to overfit the majority class:

- **Test accuracy = 0.999**, but it never predicts a fire.
- **Recall = 0**, **Precision = 0**, **PR-AUC = 0.006**, only slightly above random chance.
- **ROC-AUC = 0.53**, nearly equivalent to random guessing.

Despite class-weighting, the extreme imbalance and large categorical one-hot landcover space cause the model to default to predicting "no fire" for all samples.

3.1.3 Linear SVM

The SVM behaves similarly to the Random Forest:

- Predicts only the negative class (no fire).
- **Recall = 0**, **Precision = 0**.
- However, **ROC-AUC (0.91)** is surprisingly strong, indicating that the decision function contains separating signal even though the thresholded classifier collapses.

This suggests that a calibrated probability model or tuned classification threshold may improve practical performance.

4 XGBoost Model Performance

The XGBoost model demonstrates a significant improvement in capturing the minority class (burned area) compared to the baseline models when evaluated on the random 30% test split. Leveraging GPU acceleration (`device="cuda"`) and utilizing a robust tree-boosting architecture, our initial baseline XGBoost model achieved a Precision-Recall AUC (PR-AUC) of 0.2272 and an ROC-AUC of 0.8621 on the test set. It reached a precision of 0.53 and a recall of 0.22 for the positive class.

Following hyperparameter optimization using Optuna, the tuned model (`final_xgb_clf`) showed remarkable gains on the test data. The test accuracy increased to 0.9995, while the ROC-AUC jumped to 0.9448 and the PR-AUC more than doubled to 0.5548. In the classification report for this tuned model, the precision for predicting fire events rose to 0.78, and recall improved to 0.43, resulting in an F1-score of 0.56.

The confusion matrix for the optimized model reveals that out of 28,843 actual fire events in the test set, the model correctly predicted 12,391 (True Positives), while misclassifying only 3,404 unburned cells as fires (False Positives). This indicates that when the tuned XGBoost model predicts a fire, it is correct nearly 80% of the time, establishing a strong discriminative ability on random splits.

4.1 Feature Importance

Gradient-boosted decision trees provide inherent feature importance scores, allowing us to interpret the primary drivers of wildfire burn probability across California. Using functionality within the XGBoost library, we ranked features based on their contribution to the model's decision splits across the boosted ensemble. These scores summarize how much each variable contributes to prediction.

4.1.1 How to Interpret XGBoost Feature Importance

It is important to note that, unlike coefficients in logistic regression, XGBoost feature importance does **not** indicate whether a variable increases or decreases wildfire probability. Instead, it reflects the **relative predictive contribution** of each feature to the model. A higher importance score means that the variable plays a larger role in improving the model's splits and overall predictive performance, but it should not be interpreted as a directional or causal effect.

4.1.2 Spatial and Temporal Drivers

The feature importance results show that **spatial and temporal variables** are among the strongest contributors to prediction. In particular, `lat`, `lon`, and `year` rank among the most important features, with `month` also contributing meaningfully. This suggests that wildfire

occurrence is strongly structured by geography, seasonality, and interannual variability. In other words, the model learns that wildfire risk is not uniform across California, but instead depends heavily on where and when an observation occurs.

4.1.3 Environmental Predictors

Among the environmental predictors, **climate and vegetation variables** remain highly influential. The transformed precipitation variable `ppt_log` and dew point temperature `tdmean` are especially important, followed by `tmax` and `vpdmax_log`. Static landscape variables such as `elevation`, `ndvi`, and `slope_log` also contribute substantially. Together, these results are consistent with the wildfire literature, which shows that moisture, heat, fuel condition, and terrain all shape whether a location is susceptible to burning.

4.1.4 Role of Land Cover

The land cover indicators appear lower in the ranking overall, although some classes such as `landcover_grassland_herbaceous` and `landcover_evergreen_forest` still provide useful predictive signal. This suggests that while broad fuel type matters, it contributes less to model performance than the combined effects of location, seasonality, climate, and topography in the current statewide setting.

4.1.5 Key Takeaway

Overall, the feature importance analysis indicates that the XGBoost model relies on a combination of **where** a grid cell is located, **when** it is observed, and the **local environmental conditions** present at that time. Because these importance scores do not provide directionality, they should be interpreted as measures of predictive relevance rather than causal effect. Directional interpretation is more appropriate for the logistic regression coefficient plot included in the appendix.

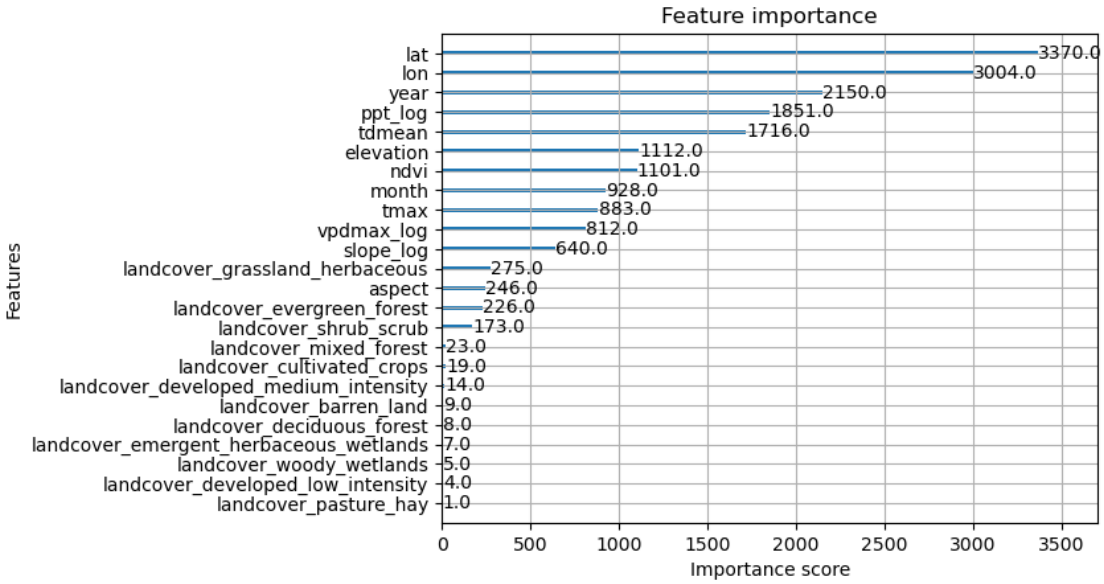


Figure 4: XGBoost feature importance scores for the tuned wildfire prediction model. These scores reflect the relative contribution of each variable to the model's tree-splitting decisions, not the direction of effect. Spatial, temporal, and climate-related predictors dominate the ranking.

5 Model Validation

Validation of the data-driven framework involves both temporal stress-testing and an assessment of its practical utility in real-world financial and safety scenarios. To rigorously assess the model's ability to generalize to unseen, future climate conditions, we evaluated the optimized XGBoost classifier on the temporally isolated validation set (years 2005–2009). The performance on this holdout set reveals important insights into the model's temporal generalizability.

While the validation accuracy remained exceptionally high at 0.9994 due to the massive number of negative samples, the minority-class metrics experienced a severe drop. The optimized model achieved an ROC-AUC of 0.8425 on the validation set, which remains reasonably strong and indicates that the overall ranking of probabilities still holds predictive signal. However, the PR-AUC plummeted to 0.0113.

The classification report and confusion matrix for the validation data highlight this performance decay: the model only correctly classified 12 out of 21,703 actual fire events, yielding a recall of near 0.00 and a precision of 0.11 (with 101 False Positives).

This significant discrepancy between the random test split and the temporal validation split suggests that the model is highly sensitive to the temporal distribution of the data. It may be overfitting to specific spatial or temporal patterns present in the training years, or the default classification threshold of 0.5 is severely misaligned for the temporal validation

set. The persistent class imbalance across different time periods underscores the need for calibrated probabilities and dynamic threshold tuning for future operational deployment.

5.1 Temporal Validation and Generalization

A temporal validation strategy was implemented to test the model's robustness against non-stationary climate patterns. The model was validated on a distinct temporal window spanning **2005–2009**. During this phase, the **PR-AUC dropped significantly to 0.0113**. This sharp decrease suggests that while the model captures patterns effectively within the primary test set, the extreme class imbalance and shifting environmental conditions over time pose a significant challenge for long-term predictive stability.

5.2 Spatial Reliability and Practical Application

Despite the challenges in temporal generalization, the spatial analysis indicates that the 800 m grid resolution remains useful for exploratory risk mapping. This burn-probability surface was developed to support the project's broader goals with Cotality. Although the weak temporal holdout results mean the model is not yet ready for deployment-oriented decision-making, it provides a proof of concept for how spatially detailed wildfire predictions could eventually inform insurance and mortgage risk metrics, as well as public safety and urban infrastructure planning within the Wildland-Urban Interface (WUI).

6 Discussion and Next Steps

Our data-driven approach marks a significant transition from traditional fire danger ratings to the prediction of actual fire activity. By fusing environmental pillars such as climate, vegetation, and historical perimeters into a monthly 800m grid, we developed a scalable statistical framework for estimating spatially detailed burn probability without relying on computationally expensive simulation pipelines, particularly in complex interfaces like the Wildland-Urban Interface (WUI).

6.1 Improving Our Current Models

While the current model provides a high-fidelity risk surface, several avenues for refinement exist to bridge the gap between proof-of-concept and operational deployment for insurance and mortgage pricing.

6.1.1 Addressing Class Imbalance

Wildfire modeling is inherently a "rare-event" problem, where the target class (burn events) represents a minute fraction of the total dataset. To improve upon our current results one could explore advanced sampling techniques such as **SMOTE (Synthetic Minority Oversampling Technique)** or cost-sensitive learning within the XGBoost framework. As noted in recent literature, optimizing for low **Expected Calibration Error (ECE)** is crucial to ensure that predicted probabilities reflect true real-world frequencies, especially when those probabilities are naturally low ($< 5\%$).

6.1.2 Model-Centric Improvements

Current results indicate that our untuned XGBoost model achieved a **test accuracy of 0.9993**, while Optuna hyperparameter tuning improved it to 0.9995. On the other hand, PR-AUC dropped to 0.0113 during temporal validation (2005–2009). This suggests the model may be overfitting to specific historical climate regimes. Future iterations will incorporate **Temporal Fold Validation** more aggressively during the hyperparameter tuning phase. Furthermore, following the findings of Di Giuseppe et al. (2025), we will investigate whether the inclusion of **lightning density** and **human-centric ignition drivers** (e.g., road density, population) can further stabilize these predictions across different time windows.

6.2 Recognizing the Limits of Current Models

It is essential to acknowledge that data-driven models are only as robust as their training inputs. Our current framework relies heavily on **Vapor Pressure Deficit (VPD)** and **maximum temperature**, which are strong predictors but can lead to over-prediction in sparsely vegetated areas if fuel abundance is not accurately captured.

7 Conclusion

This project developed a statewide, data-driven wildfire burn-probability model for California using publicly available environmental data. Using XGBoost, the model captured nonlinear relationships among climate, vegetation, terrain, and time that simpler baseline approaches may miss. In partnership with Cotality, this work shows the promise of a scalable framework for producing more spatially detailed wildfire risk information relevant to insurance, mortgage risk, and resilience planning. However, the weak temporal holdout performance also makes clear that future generalization remains the main limitation of the current approach. Overall, the project suggests that directly modeling wildfire occurrence, rather than relying only on static fire-danger indicators, is a promising direction for more adaptive and operationally useful risk assessment.

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Appendices

A.1 Training Details A1
A.2 Additional Figures A2
A.3 Contributions A4

A.1 Training Details

A.1.1 Data Splitting

- Held out all observations from **2005–2009** as a temporally separated validation set.
- Remaining years (2000–2004, 2010–2024) were split into:
  - **70% training set**
  - **30% test set**
- Split performed using `train_test_split` **without stratification**.
- StandardScaler fit on training data and applied to test + validation sets for LR and SVM.

A.1.2 Model Parameters

- **Logistic Regression**
  - `penalty = 'l2'`
  - `max_iter = 1000`
  - `class_weight = 'balanced'`
  - `solver = 'lbfgs'`
- **Random Forest**
  - `n_estimators = 200`
  - `max_depth = 15`
  - `max_features = 'sqrt'`
  - `class_weight = 'balanced_subsample'`
  - `n_jobs = -1`
- **Linear SVM**
  - `LinearSVC`
  - `dual = False`
  - `max_iter = 1000`
  - No class weights used in baseline model.

A.2 Additional Figures

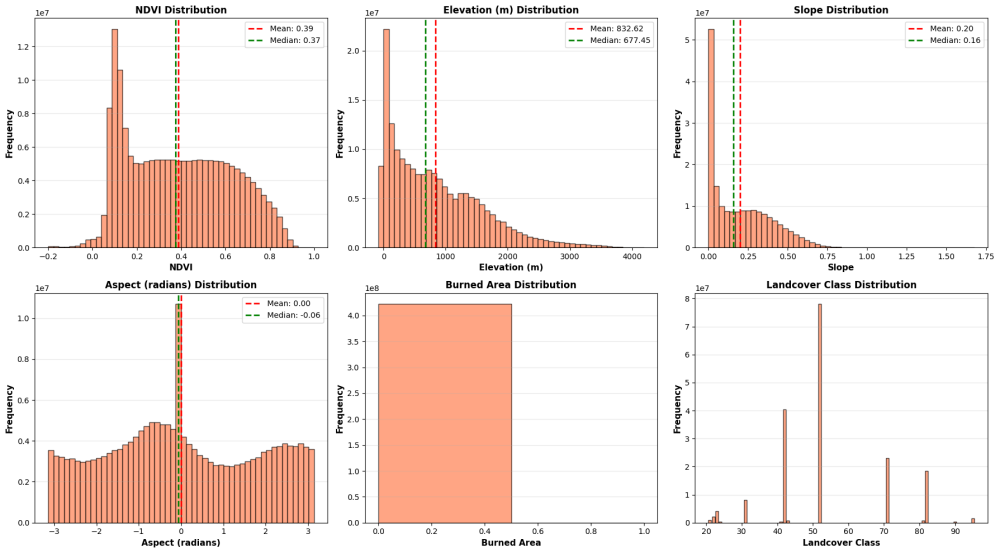

Figure A 1: Additional distributions for NDVI, elevation, slope, aspect, burned area, and land cover across the California dataset. These plots provide supporting evidence for skewness, class imbalance, and the categorical structure of land cover.

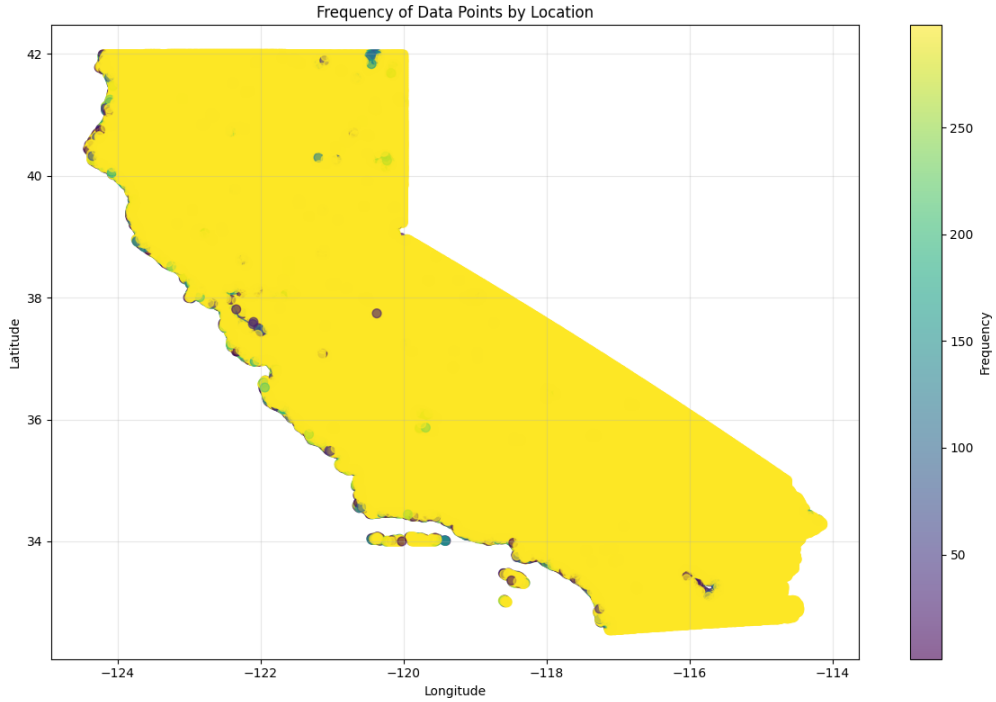

Figure A 2: Frequency of available observations by spatial location across the California domain. Most locations have high temporal coverage, with lower counts concentrated around boundary and masked regions.

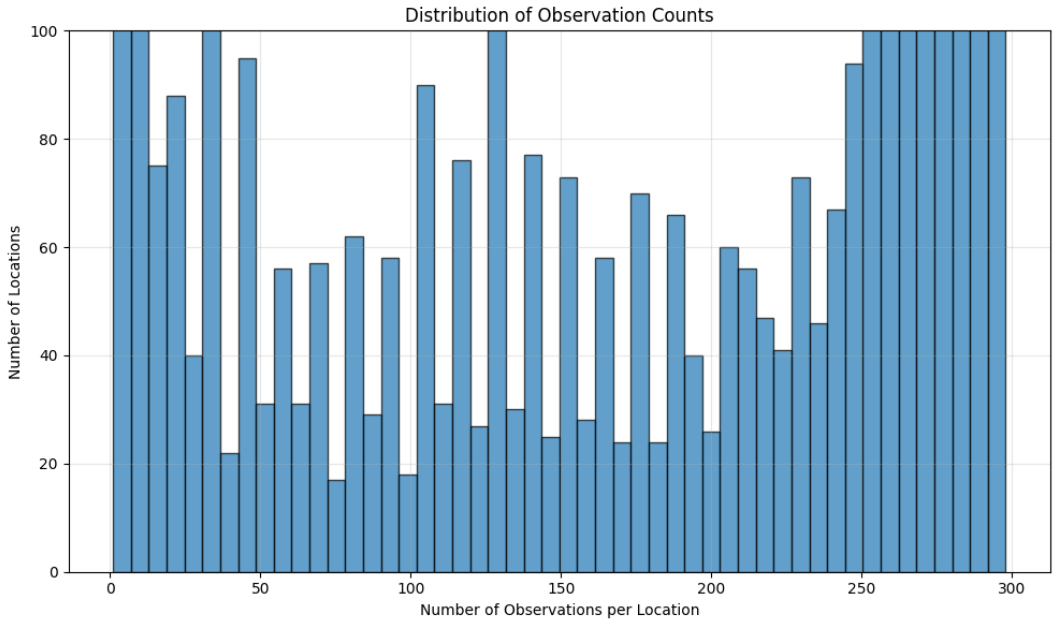

Figure A 3: Distribution of observation counts per location. This figure summarizes the temporal completeness of each grid cell after filtering and merging the input datasets.

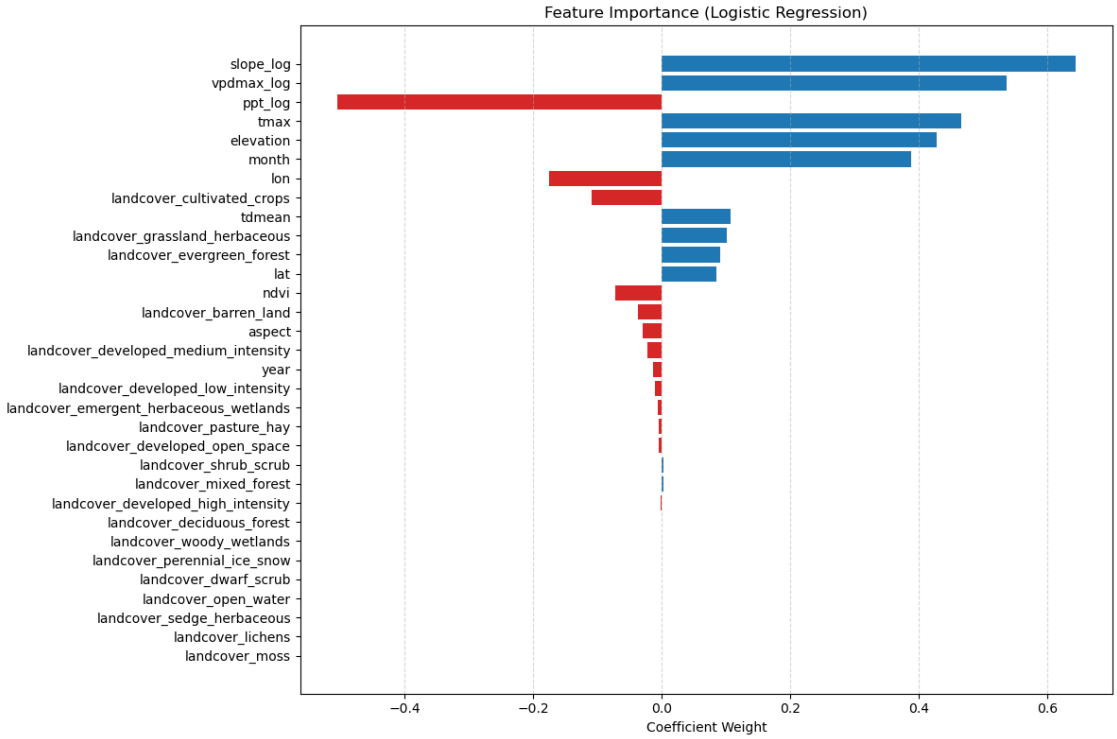

Figure A 4: Logistic regression coefficient weights for the baseline linear model. Unlike XGBoost importance, these coefficients provide directional information, indicating whether a feature is associated with higher or lower predicted wildfire probability under the linear specification.

A.3 Contributions

- **Stuti Verma:** Led data acquisition and large-scale preprocessing of statewide California datasets (PRISM, MODIS NDVI, NLCD, MTBS), while also driving much of the project planning and coordination. Contributed heavily to documentation of the data pipeline, ensured the reproducibility of preprocessing steps, and led report writing efforts.
- **Gahn Wuwong:** Contributed to technical documentation and visualization development for the appendix. Designed and implemented baseline models (Logistic Regression, Random Forest, Linear SVM). Led the development of the XGBoost framework with GPU acceleration and Optuna hyperparameter tuning. Contributed to final report writing, results interpretation, and overall project coordination.
- **Lacha Barton-Gluzman:** Conducted exploratory data analysis (EDA), including missing data diagnostics, distribution analysis, and multicollinearity assessment. Implemented feature engineering techniques such as lagging, log transformations, and PCA. Played a key role in project planning, modeling strategy decisions, and the integration of model outputs into the final narrative. Served as the primary point of contact with the industry mentor, facilitating feedback integration. Contributed to final report writing, results interpretation, and overall project coordination.
- **Diego Arevalo Fernandez:** Developed the validation framework, including temporal holdout design and train/test splitting strategy. Analyzed model performance metrics and interpreted tradeoffs between ROC-AUC and PR-AUC. Contributed to final report writing, results interpretation, and overall project coordination. Supported collaborative planning of modeling experiments and iterative refinement.